Mutual Fund Analytics Capstone Project

Bluestock Fintech

Data Analyst Internship Capstone Project

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B.Sc. Information Technology
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Project Overview

- The Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, and portfolio risk using Python, SQL, SQLite, and Power BI. The project covers the complete analytics workflow from data ingestion and cleaning to advanced financial metrics and interactive dashboards, enabling data-driven investment analysis.
- **Technology Stack**

- Python
- Pandas & NumPy
- SQLite
- SQL
- Jupyter Notebook
- Power BI
- Git & GitHub

Project Objectives

- Analyze mutual fund performance using historical NAV data.
- Build a robust ETL pipeline for data ingestion, cleaning, and processing.
- Store cleaned data efficiently in a SQLite database.
- Perform Exploratory Data Analysis (EDA) to identify investment trends.
- Calculate key financial performance metrics such as CAGR, Sharpe Ratio, Alpha, Beta, Sortino Ratio, Maximum Drawdown, VaR, and CVaR.
- Develop an interactive Power BI dashboard for visualization.
- Build a rule-based mutual fund recommendation system.

Dataset Overview

The project uses ten primary datasets covering mutual fund schemes, NAV history, investor transactions, portfolio holdings, SIP inflows, benchmark indices, and industry statistics.

Dataset	Records	Purpose
Fund Master	40	Mutual fund scheme information
NAV History	46,000	Historical NAV prices
AUM by Fund House	90	Assets under Management
Monthly SIP Inflows	48	SIP market trends
Category Inflows	144	Category-wise fund inflows
Industry Folio Count	21	Industry folio statistics
Scheme Performance	40	Performance metrics
Investor Transactions	32,778	Investor behaviour analysis
Portfolio Holdings	322	Portfolio composition
Benchmark Indices	8,050	Market benchmark comparison

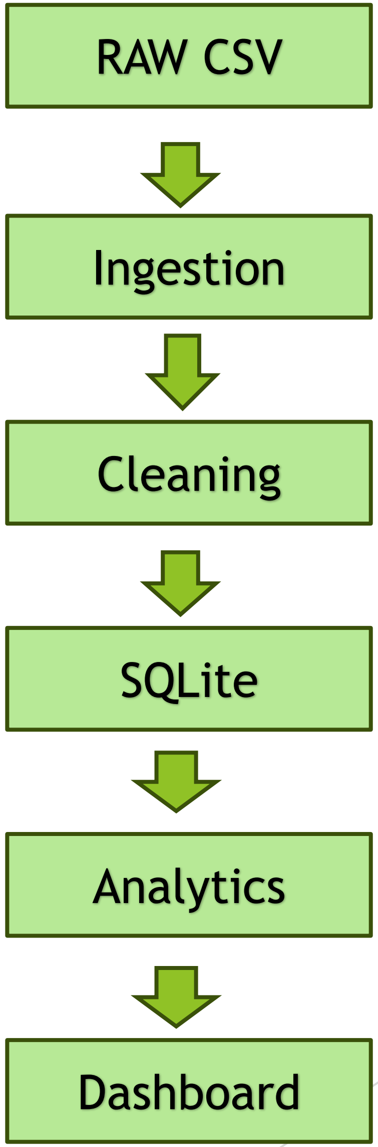
ETL Pipeline

The ETL (Extract, Transform, Load) pipeline automates the complete data processing workflow. Raw mutual fund datasets are validated, cleaned, transformed, and finally loaded into a SQLite database for analysis and dashboard development.

Key Features

- Automated ingestion
- Data cleaning
- Missing value handling
- SQLite loading
- Error validation

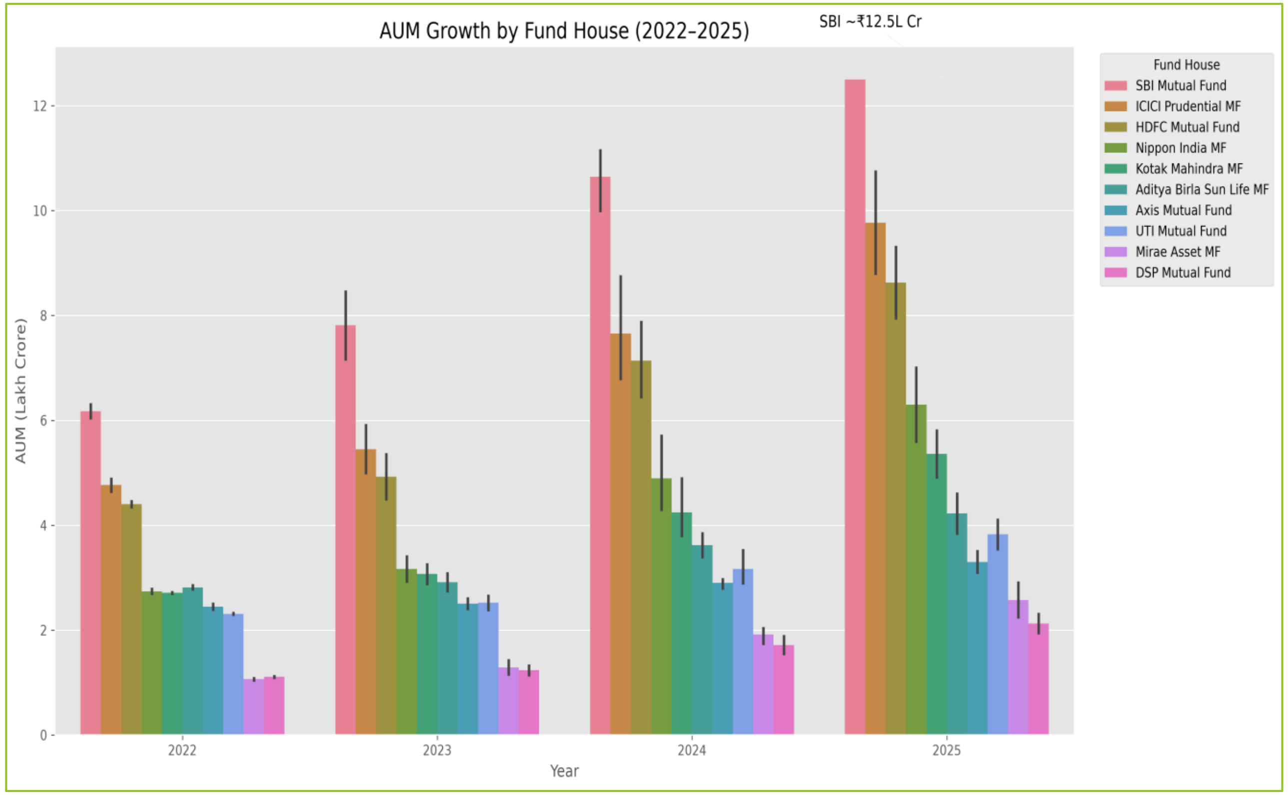
ETL Workflow



Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify investment trends, fund growth patterns, investor behaviour, and category-wise inflows using Python and Pandas

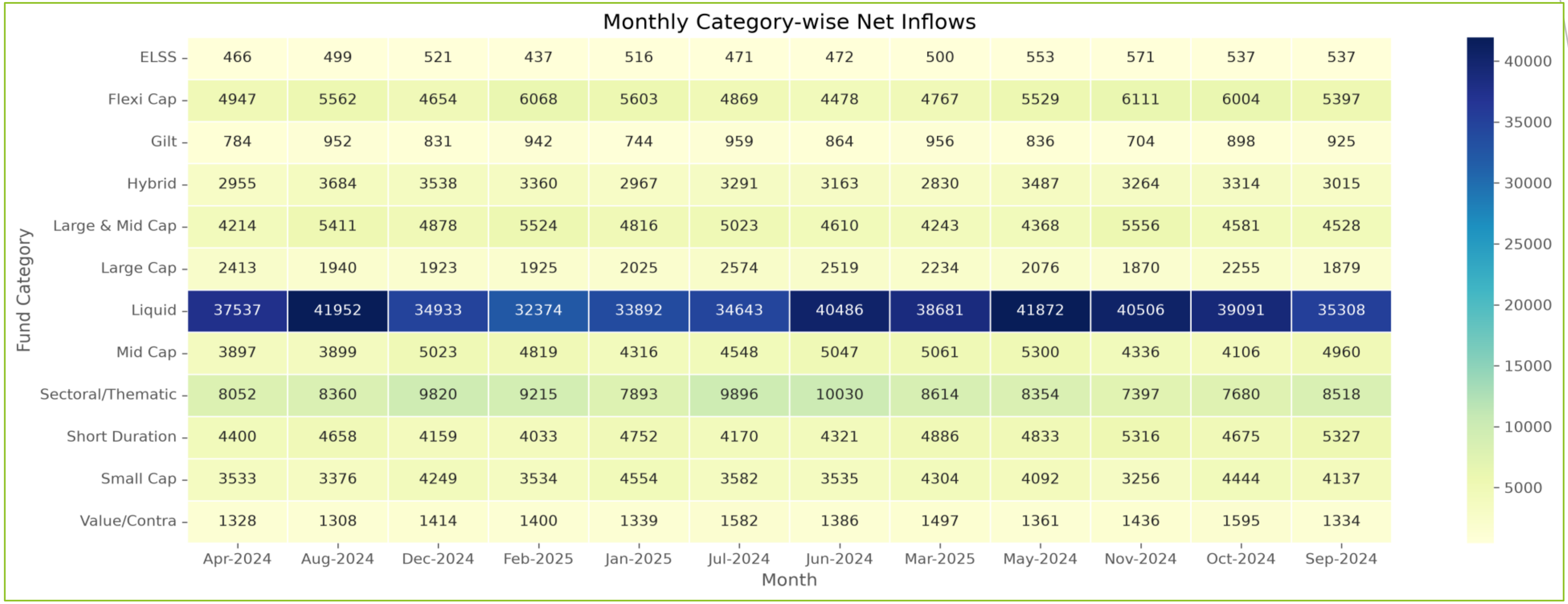
AUM Growth by Fund House



Key Insights

- Assets Under Management (AUM) increased steadily from **2022 to 2025**, indicating consistent industry growth.
- Large fund houses such as **SBI Mutual Fund, HDFC Mutual Fund, and ICICI Prudential MF** maintained higher AUM compared with others.
- The upward trend reflects increasing investor confidence and continued inflows into mutual funds.

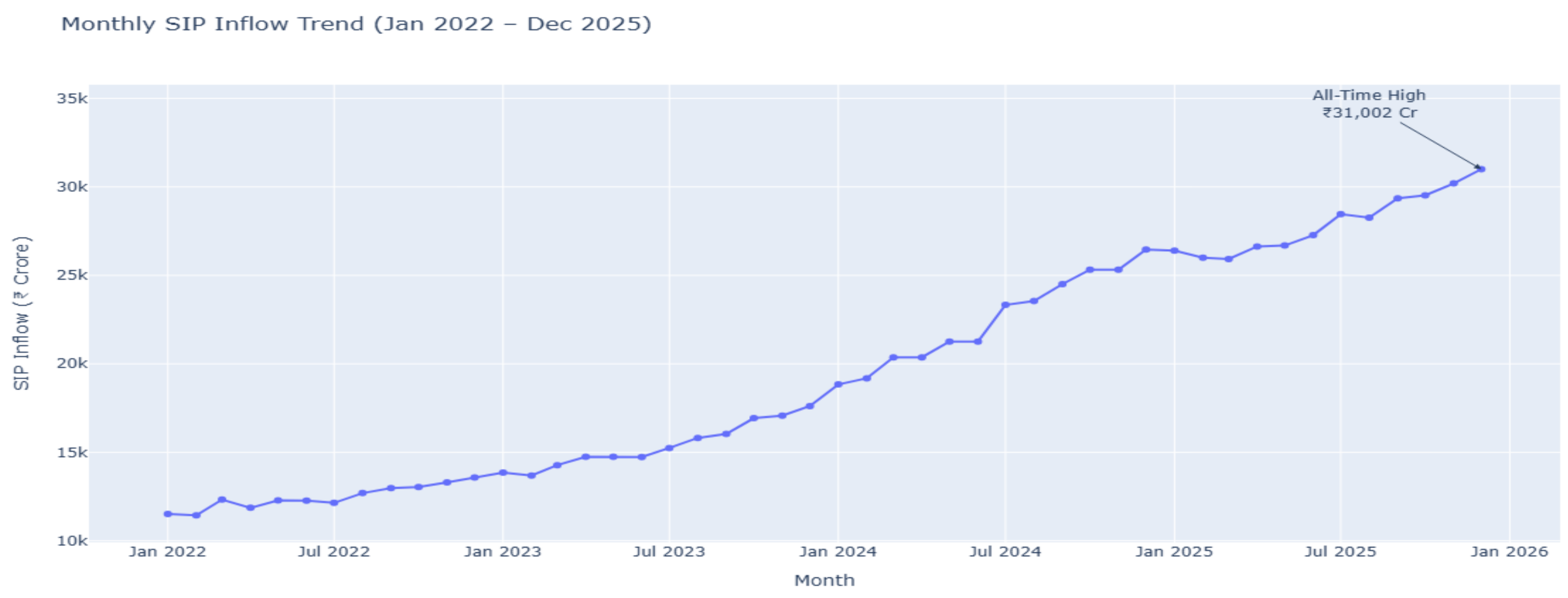
Category-wise Net Inflow Heatmap



Key Insights

- Flexi Cap, Mid Cap, and Large & Mid Cap categories recorded comparatively higher net inflows across multiple months.
- Category-wise inflows varied over time, showing changing investor preferences.
- Most categories maintained positive inflows during the analysis period, indicating healthy market participation.

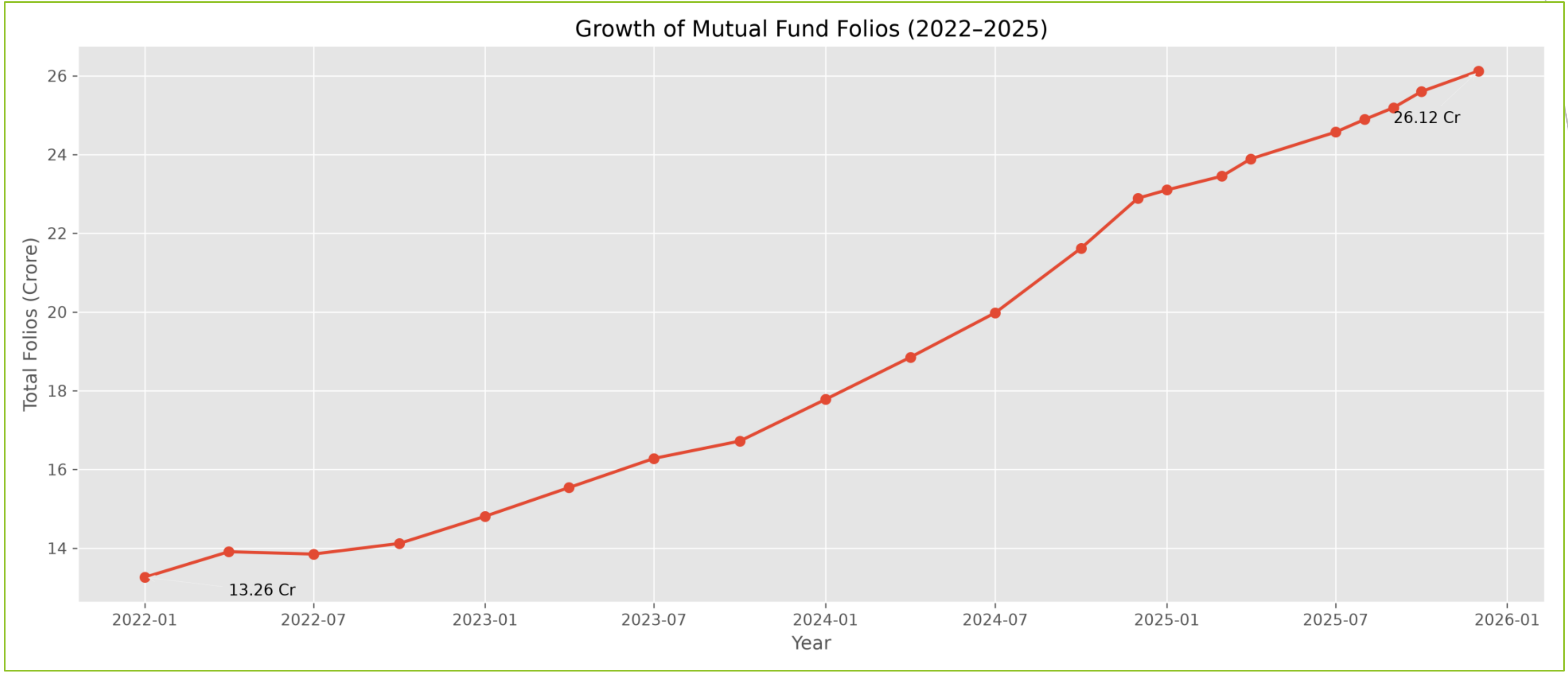
Monthly SIP Inflow Trend



Key Insights

- Monthly SIP collections showed an overall upward trend during the study period.
- Regular SIP investments indicate growing participation from retail investors.
- Consistent SIP inflows demonstrate increasing long-term investment discipline.

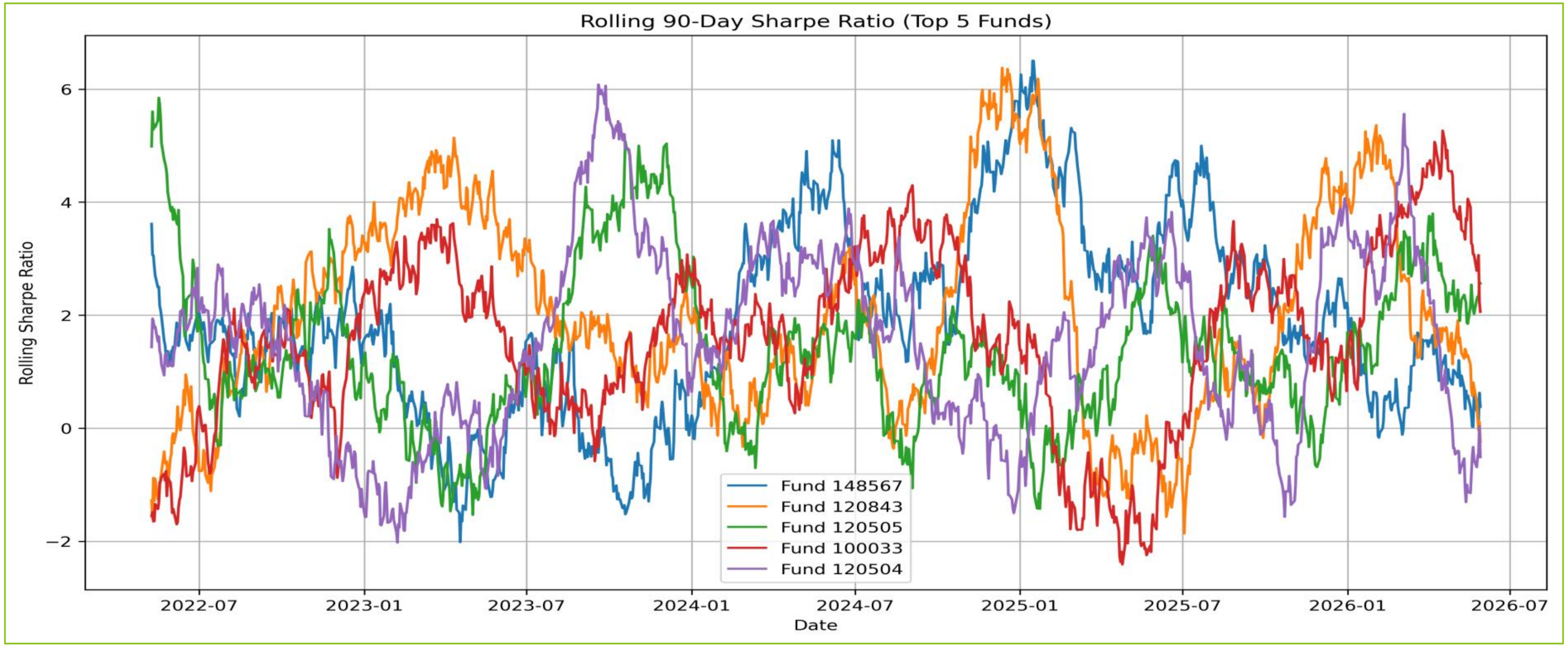
Industry Folio Growth



Key Insights

- Total mutual fund folios increased steadily between **2022 and 2025**.
- The rise in folios indicates expanding retail investor participation in mutual funds.
- Continuous folio growth reflects increasing awareness and adoption of mutual fund investments.

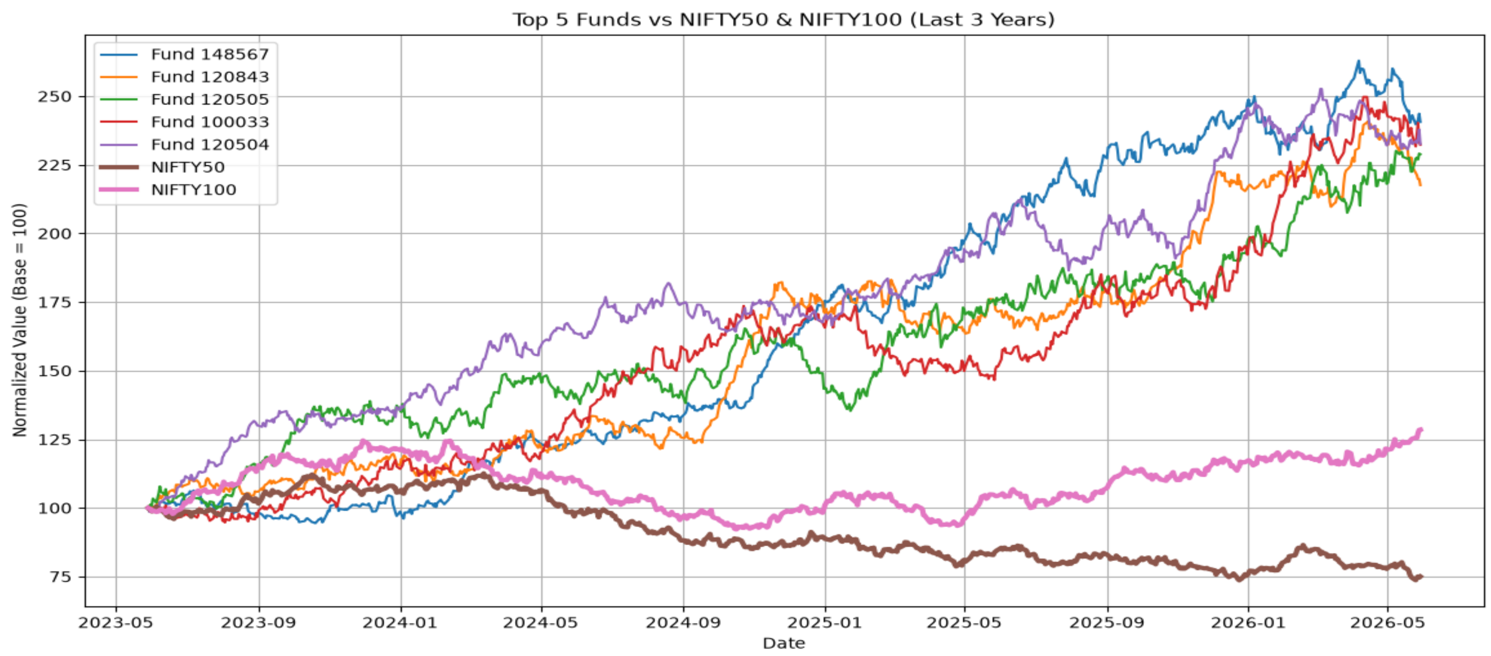
Performance Analytics



Key Insights

- Rolling Sharpe Ratio fluctuated over time, reflecting changing market conditions.
- Higher Sharpe values indicate better risk-adjusted returns during those periods.
- The metric helps compare fund performance while considering both return and volatility.

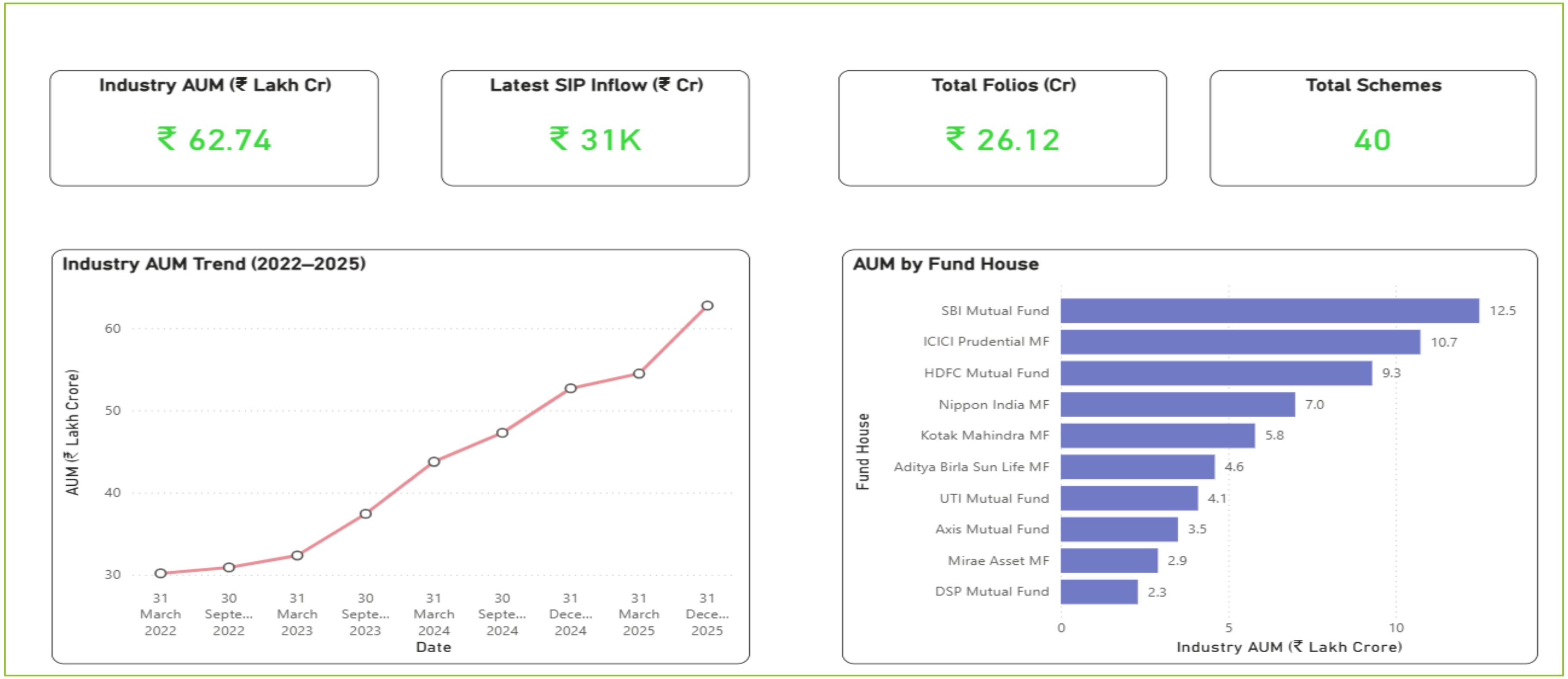
Advanced Analytics



Key Insights

- Benchmark comparison helps measure whether funds outperform or underperform the market.
- VaR estimates the maximum expected loss at a chosen confidence level.
- CVaR measures the average loss beyond the VaR threshold, providing deeper downside risk analysis.

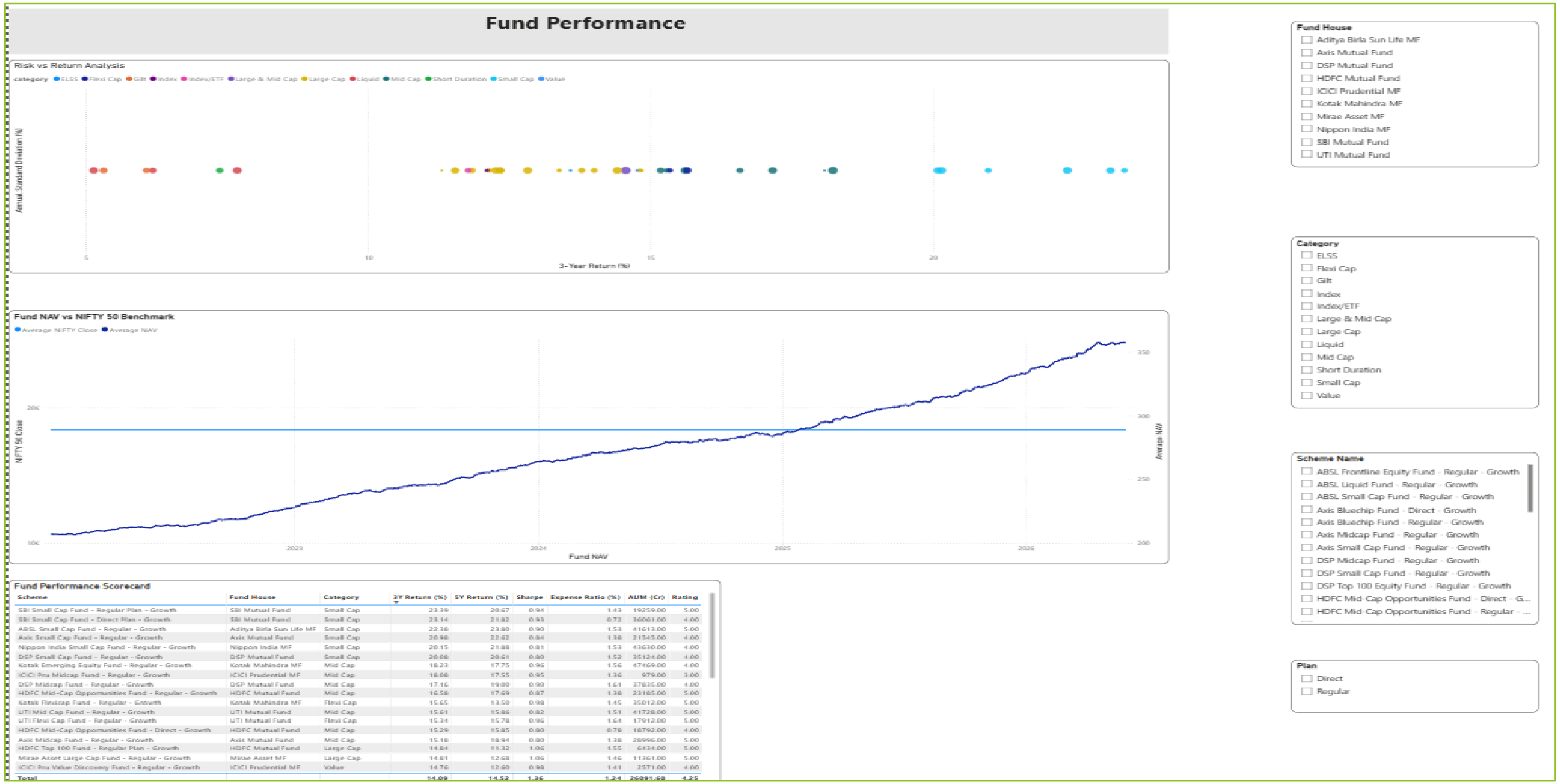
Industry Overview



Key Features

- Presents key mutual fund industry KPIs in a single view.
- Shows AUM growth and category-level trends.
- Interactive slicers enable dynamic data exploration.

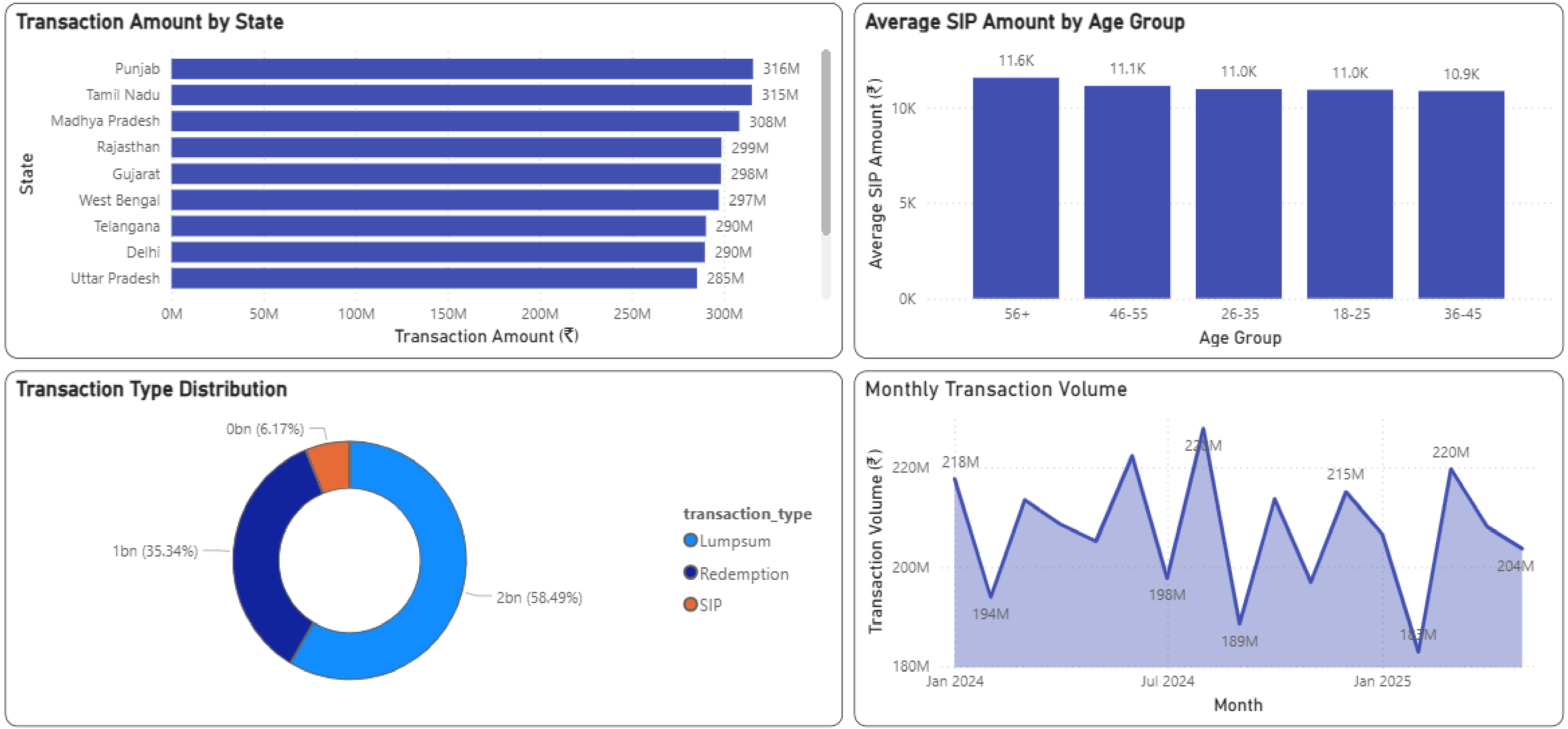
Fund Performance



Key Features

- Compares fund performance using return and risk metrics.
- Highlights top-performing schemes across categories.
- Enables performance comparison through interactive visuals.

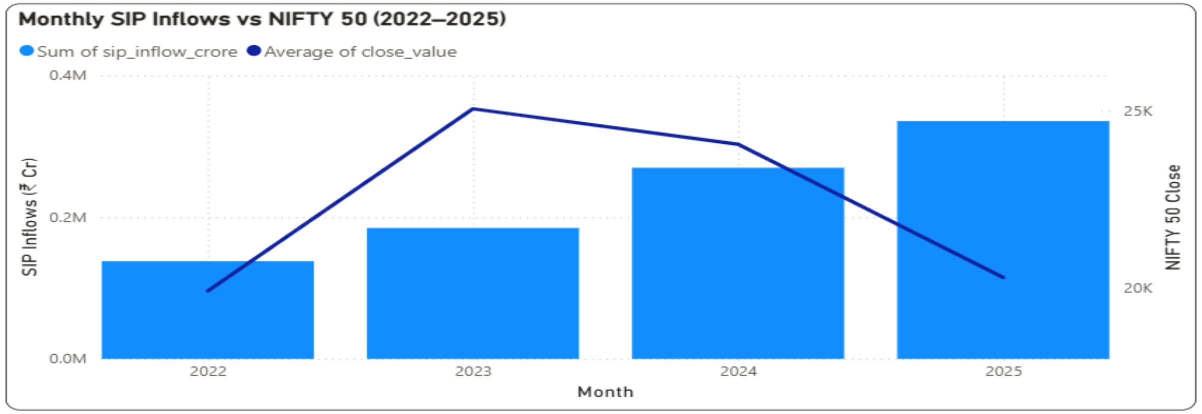
Investor Analytics



Key Features

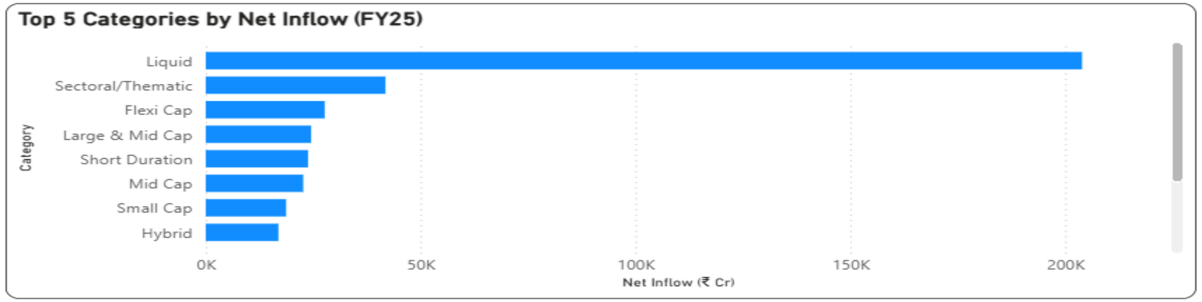
- Analyzes investor demographics and transaction patterns.
- Displays participation across different regions and age groups.
- Helps understand investor behaviour using interactive filters.

SIP & Market Trends



Category-wise Net Inflow Heatmap

Category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Total	**706618**	**225621**	**932239**



Key Features

- Visualizes monthly SIP inflows and market activity.
- Tracks industry folio growth over time.
- Supports trend analysis for long-term investment behaviour.

Key Findings

- ✓ Successfully developed an end-to-end Mutual Fund Analytics solution using Python, SQLite, and Power BI.
- ✓ Automated the ETL pipeline for data ingestion, cleaning, validation, and database loading.
- ✓ Performed Exploratory Data Analysis (EDA) to identify fund growth, SIP trends, and investor participation.
- ✓ Calculated financial performance metrics such as CAGR, Sharpe Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR.
- ✓ Built an interactive Power BI dashboard with multiple analytical views for fund performance and investor insights.
- ✓ Implemented a rule-based mutual fund recommendation system to support investment decisions.

Conclusion

- Successfully transformed raw mutual fund datasets into meaningful business insights.
- Integrated ETL, data analytics, financial analysis, and interactive dashboards into a single workflow.
- The project demonstrates practical application of data analytics techniques for investment analysis.

Thank You

Thank You!

Questions & Discussion

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